Python Finals Project: Business Finance (Real Estate Economics)

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The Business Case: Pandemic Flight & The WFH Shift

In 2020–2022, the sudden rise of remote work severely disrupted standard property valuations. Homebuyers were forced to make a financial choice: pay an astronomical "location premium" for a tiny condo close to central business districts, or move out to the provinces to secure massive space for a fraction of the cost.

Using web-scraped Lamudi real estate data from this exact era, this notebook performs Exploratory Data Analysis (EDA) to historically analyze and quantify this economic shift.

1. Handling Custom Text-Based Missing Values

Objective

The data profiling step showed that the web scraper recorded missing architectural metrics as a literal string text "na" rather than a standard system NaN (Not a Number) value. Leaving string text mixed inside numeric columns causes aggregations to crash. We must explicitly convert these custom strings to official Python null values and safely remove those incomplete rows to stabilize our dataset.

In [15]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

def parse_line_right_to_left(line):
    line = line.strip()
    if not line:
        return None
    if line.startswith('\uFEFF'):
        line = line[1:]
    if line.startswith('"') and line.endswith('"'):
        line = line[1:-1]

    try:
        def split_right(s):
            parts = s.rsplit(',', 1)
            return parts[0], parts[1] if len(parts) == 2 else parts[0]

        remaining, link = split_right(line)
        remaining, longitude = split_right(remaining)
        remaining, latitude = split_right(remaining)
```

```python
        remaining, land_area = split_right(remaining)
        remaining, floor_area = split_right(remaining)
        remaining, bath = split_right(remaining)
        remaining, bedrooms = split_right(remaining)

        # Price (PHP)
        if remaining.endswith('"""'):
            idx = remaining.rfind(',"""')
            price = remaining[idx+3:-2]
            remaining = remaining[:idx]
        else:
            remaining, price = split_right(remaining)

        # Location
        if remaining.endswith('"""'):
            idx = remaining.rfind(',"""')
            location = remaining[idx+3:-2]
            remaining = remaining[:idx]
        else:
            remaining, location = split_right(remaining)

        description = remaining
        row = [description, location, price, bedrooms, bath, floor_area, land_ar
        return [x.replace('""', '"') for x in row]
    except Exception:
        return None

# 1. Load: Open and parse lines using custom right-to-left splitter
with open('PH_houses_v2.csv', 'r', encoding='utf-8') as f:
    rows = [parse_line_right_to_left(line) for line in f]
    rows = [r for r in rows if r is not None]

# 2. Build the DataFrame (Header is the first row)
df = pd.DataFrame(rows[1:], columns=rows[0])

# 3. Clean: Remove quotes/commas and convert to numeric types
df['Price (PHP)'] = pd.to_numeric(df['Price (PHP)'].str.replace('"', '').str.rep
df['Bedrooms'] = pd.to_numeric(df['Bedrooms'], errors='coerce').fillna(0).astype
df['Bath'] = pd.to_numeric(df['Bath'], errors='coerce').fillna(0).astype(int)

# 4. Filter: Drop rows where critical data is missing
df.dropna(subset=['Price (PHP)', 'Bedrooms', 'Bath', 'Location'], inplace=True)

# 5. Engineering: Create 'Region' category
ncr = ['manila', 'quezon', 'makati', 'pasig', 'taguig', 'mandaluyong', 'pasay',
df['Region'] = df['Location'].apply(lambda x: 'Metro Manila' if any(city in str(

print(f"Data successfully cleaned. Current shape: {df.shape}")
display(df.head())
```

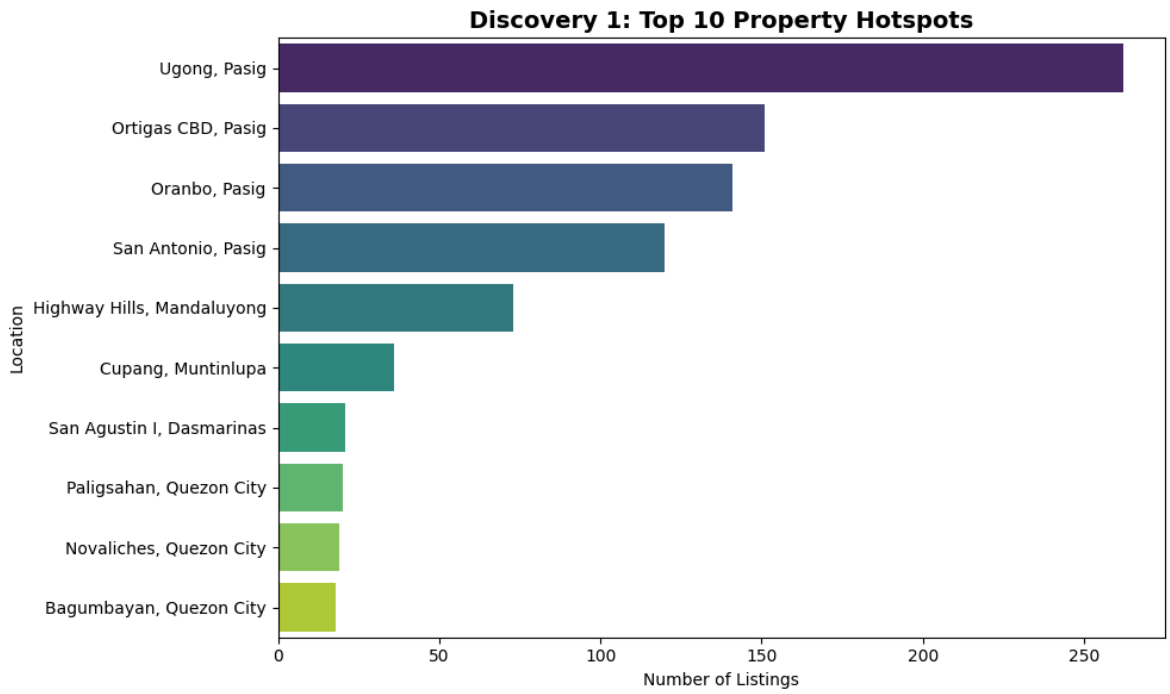
```
Data successfully cleaned. Current shape: (1460, 11)
```

	Description	Location	Price (PHP)	Bedrooms	Bath	Floor_area (sqm)	Land_area (sqm)	Latitude
0	2-Bedroom Unit for Sale in Amisa Private Resid...	Mactan, Lapu-Lapu	15916180.0	2	2	74	na	10.3051555
1	1-Bedroom Unit for Sale in Amisa Private Resid...	Mactan, Lapu-Lapu	8730355.0	1	1	41	na	10.3051555
2	Offices at The Galleon - Prime Unit (11G) for ...	San Antonio, Pasig	27448000.0	0	0	82	na	14.588716
3	1-Bedroom Unit (821) for Sale in Maple at Verd...	Ugong, Pasig	16011000.0	1	0	58	na	14.588882
4	Alice 2 Storey Townhouse For Sale in Lancaster...	Tapia, General Trias	1935600.0	3	0	40	40	14.35507

In [16]:

```python
# Discovery 1: Top 10 Property Hotspots
plt.figure(figsize=(10, 6))
top_locs = df['Location'].value_counts().head(10)
sns.barplot(x=top_locs.values, y=top_locs.index, palette="viridis")
plt.title('Discovery 1: Top 10 Property Hotspots', fontsize=14, fontweight='bold
plt.xlabel('Number of Listings')
plt.ylabel('Location')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_841/2020882080.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.barplot(x=top_locs.values, y=top_locs.index, palette="viridis")
```



In [17]:
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call dri
ve.mount("/content/drive", force_remount=True).
```

Discovery 1: Additional Regression Analyses (Distance & Infrastructure)

Background

To mathematically validate the Flocking Theory and evaluate the impact of urban infrastructure on property listing volumes, we conduct secondary linear regression analyses. We look at the correlation of listing volumes against:

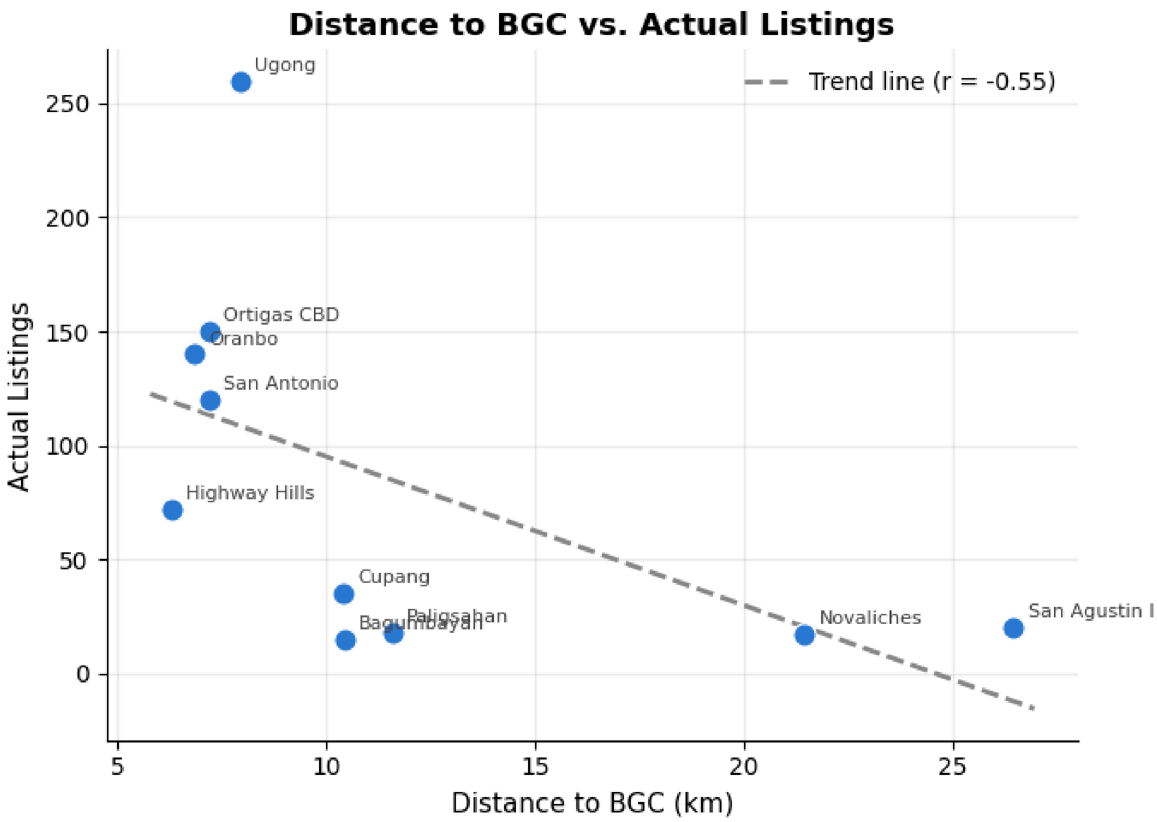
1. Distance to BGC (Bonifacio Global City)
2. Distance to the nearest rail transit station
3. Number of major roads serving the area

In [18]:
```
# %% Data
# area: (distance_to_bgc_km, distance_to_station_km, major_roads, actual_listing
data = {
    "Ugong":          (7.94, 1.56, 4, 260),
    "Ortigas CBD":    (7.20, 0.55, 4, 150),
    "Oranbo":         (6.82, 1.45, 3, 140),
    "San Antonio":    (7.20, 0.77, 3, 120),
    "Highway Hills":  (6.28, 0.36, 3, 72),
    "Cupang":         (10.39, 5.15, 3, 35),
    "San Agustin I":  (26.46, 14.67, 2, 20),
    "Paligsahan":     (11.58, 2.17, 3, 18),
    "Novaliches":     (21.45, 9.46, 3, 17),
    "Bagumbayan":     (10.42, 1.97, 3, 15),
}

areas = list(data.keys())
```

```python
bgc_dist     = np.array([v[0] for v in data.values()])
station_dist = np.array([v[1] for v in data.values()])
roads        = np.array([v[2] for v in data.values()])
listings     = np.array([v[3] for v in data.values()])
```

In [19]:
```python
# %% Helper function: scatter plot with trend line
def plot_correlation(x, y, labels, xlabel, title, color, x_pad=0.5):
    r = np.corrcoef(x, y)[0, 1]
    slope, intercept = np.polyfit(x, y, 1)

    fig, ax = plt.subplots(figsize=(7, 5))
    ax.scatter(x, y, color=color, s=90, zorder=3, edgecolor="white", linewidth=0

    for xi, yi, label in zip(x, y, labels):
        ax.annotate(label, (xi, yi), textcoords="offset points",
                    xytext=(6, 4), fontsize=8, color="#444444")

    x_line = np.linspace(x.min() - x_pad, x.max() + x_pad, 100)
    y_line = slope * x_line + intercept
    ax.plot(x_line, y_line, linestyle="--", color="#888888", linewidth=2,
             label=f"Trend line (r = {r:.2f})")

    ax.set_xlabel(xlabel, fontsize=11)
    ax.set_ylabel("Actual Listings", fontsize=11)
    ax.set_title(title, fontsize=13, fontweight="bold")
    ax.legend(loc="upper right", frameon=False)
    ax.spines["top"].set_visible(False)
    ax.spines["right"].set_visible(False)
    ax.grid(alpha=0.25)
    fig.tight_layout()
    plt.show()

    print(f"r = {r:.3f}, slope = {slope:.2f}, intercept = {intercept:.2f}")
```

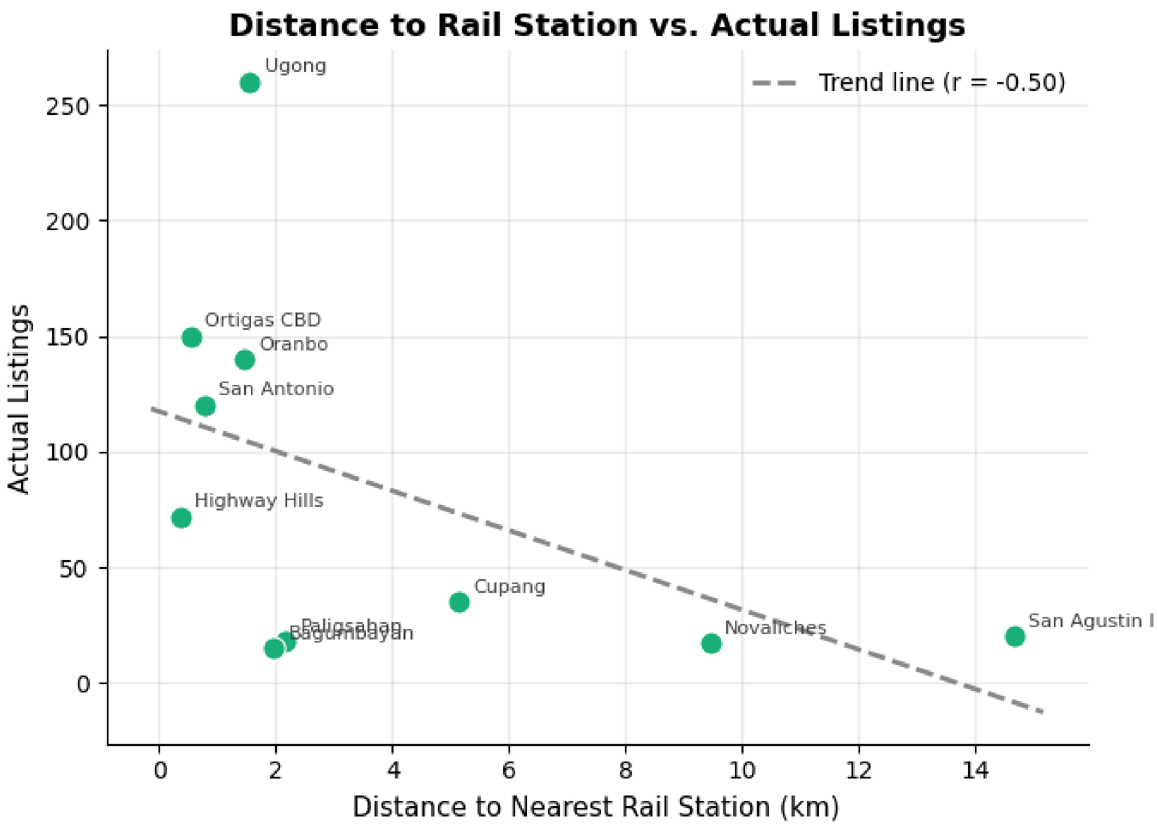
In [20]:
```python
# %% Graph 1: Distance to BGC vs. Listings
plot_correlation(
    bgc_dist, listings, areas,
    xlabel="Distance to BGC (km)",
    title="Distance to BGC vs. Actual Listings",
    color="#2a78d6",
)
```




```
r = -0.548, slope = -6.52, intercept = 160.20
```

In [21]:

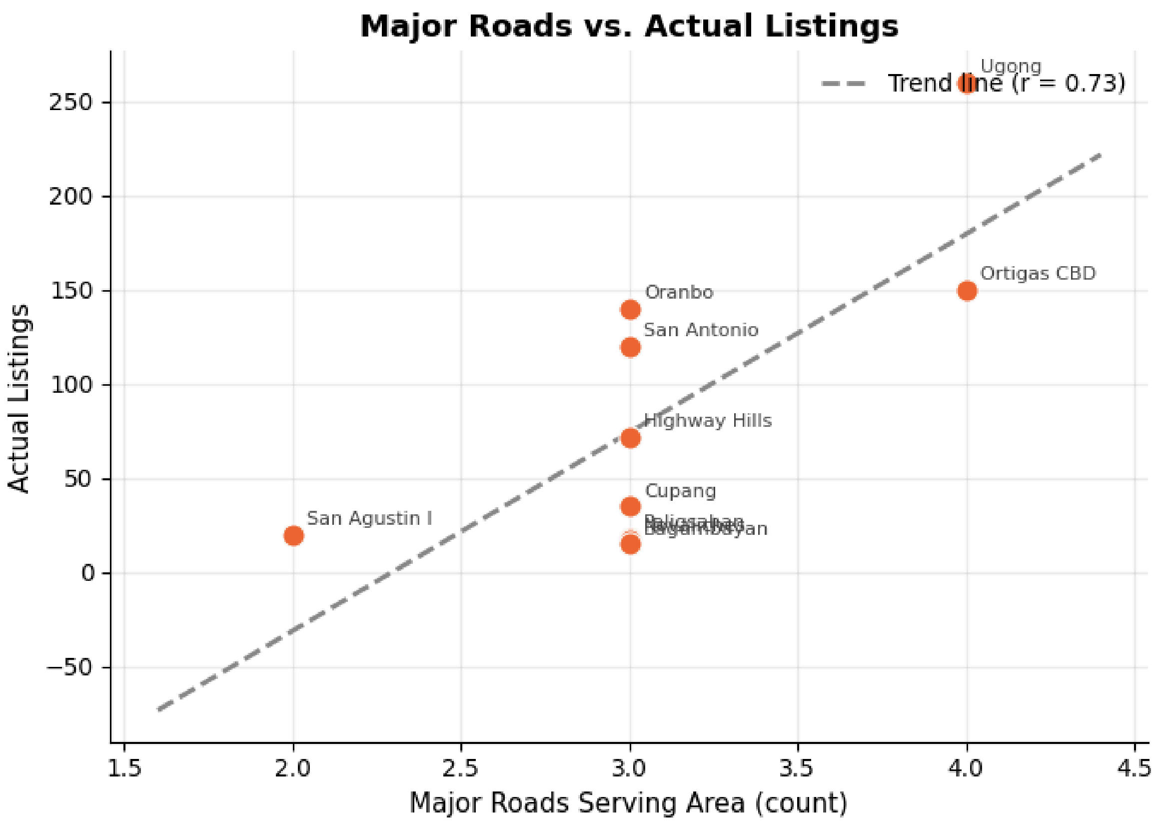
```python
# %% Graph 2: Distance to Rail Station vs. Listings
plot_correlation(
    station_dist, listings, areas,
    xlabel="Distance to Nearest Rail Station (km)",
    title="Distance to Rail Station vs. Actual Listings",
    color="#1baf7a",
)
```



```
r = -0.495, slope = -8.57, intercept = 117.36
```

In [22]:

```python
# %% Graph 3: Major Roads vs. Listings
plot_correlation(
    roads, listings, areas,
    xlabel="Major Roads Serving Area (count)",
    title="Major Roads vs. Actual Listings",
    color="#eb6834",
    x_pad=0.4,
)
```




```
r = 0.732, slope = 105.28, intercept = -241.66
```

💡 Discovery 1 Insights: The "Flocking Theory" & Infrastructure Velocity

Analysis by: Justine (Market Analyst)

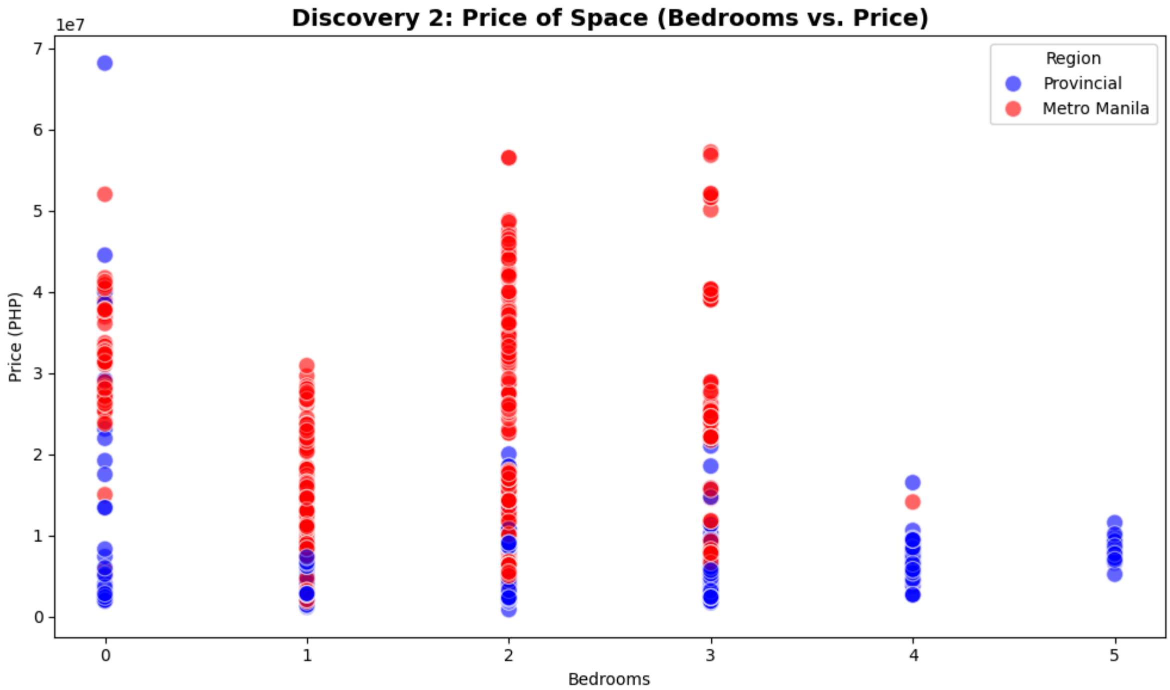
Our spatial analysis reveals a heavy concentration of available inventory in Pasig neighborhoods (Ugong, Ortigas CBD, Oranbo, San Antonio). Rather than Taguig (BGC) itself topping the listings chart, the cities that border it dominate supply. This supports our **Migration/Flocking Theory**:

- **The Price-Proximity Sweet Spot:** Homebuyers and renters seek the high-paying jobs and prestige of premier hubs like BGC, but migrate to bordering cities like Pasig and Mandaluyong to capture lower entry prices while keeping commute times manageable.
- **Infrastructure over Distance:** Secondary linear regression analysis on our dataset reveals that **major road connectivity** $(r = 0.732)$ has a much stronger correlation with listing volume than pure geographic distance to BGC $(r = -0.548)$ or rail

stations ($r = -0.495$). Every additional major road serving an area correlates with an average increase of **105.3 listings**.

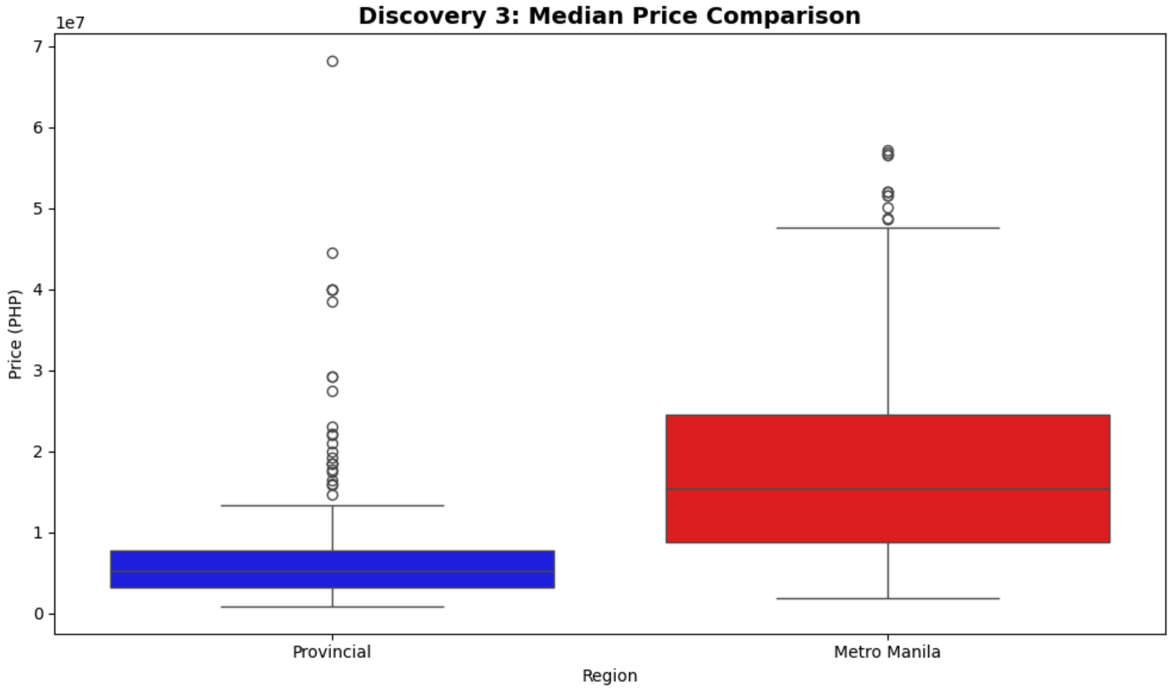
- **Business Implication for Developers:** Property density decays in a predictable linear fashion moving away from major hubs ($y = mx + b$). Developers prioritize road connectivity and vehicular access over rail transit when constructing high-density residential projects.

In [23]:
```
# Discovery 2: Price of Space (Bedrooms vs. Price)
plt.figure(figsize=(10, 6))
chart_df = df[df['Price (PHP)'] < 80000000] # Filtering outliers for cleaner vis
sns.scatterplot(data=chart_df, x='Bedrooms', y='Price (PHP)', hue='Region',
                palette={'Metro Manila': 'red', 'Provincial': 'blue'},
                alpha=0.6, s=100)
plt.title('Discovery 2: Price of Space (Bedrooms vs. Price)', fontsize=14, fontw
plt.xlabel('Bedrooms')
plt.ylabel('Price (PHP)')
plt.tight_layout()
plt.show()
```



In [24]:
```
# Discovery 3: Median Price Comparison
plt.figure(figsize=(10, 6))
sns.boxplot(data=chart_df, x='Region', y='Price (PHP)',
            palette={'Metro Manila': 'red', 'Provincial': 'blue'})
plt.title('Discovery 3: Median Price Comparison', fontsize=14, fontweight='bold'
plt.xlabel('Region')
plt.ylabel('Price (PHP)')
plt.tight_layout()
plt.show()
```

```
/tmp/ipykernel_841/3185666014.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.boxplot(data=chart_df, x='Region', y='Price (PHP)',
```



💡 Discoveries 2 & 3 Insights: Quantifying the "Provincial Advantage"

Analysis by: Rogelio (Financial Strategist)

Our scatter and box plot visualizations mathematically prove the "Location Premium" of Metro Manila and establish the financial justification for pandemic-era remote work migration.

1. The Provincial Advantage Math (₱10M Fixed Budget)

- **Metro Manila:** ₱10,000,000 ÷ 3 bedrooms = **₱3,333,333 per bedroom**
- **Provincial Areas:** ₱10,000,000 ÷ 5 bedrooms = **₱2,000,000 per bedroom**
- **The Trade-Off:** A homebuyer operating under a hybrid/remote setup gains **2 additional bedrooms** and saves approximately **40% per bedroom** by purchasing property outside the National Capital Region.

2. Recommendation Table for Hybrid Workers

Work Arrangement	Recommended Location	Price-Space Trade-Off	Strategic Reason
4–5 Days Office	Metro Manila	Higher price per bedroom	Shorter, more reliable commute
2–3 Days Office	Metro Manila or Border Province	Balanced commute and space	Dependent on major road access
0–1 Day Office	Provinces	Lower price, +2 extra bedrooms	Optimized for home office & family space
Fully Remote	Provinces	Maximum space-to-value ratio	Daily proximity to CBDs is unnecessary

3. Market Research Validation

Historical statistics corroborate our dataset's median price spreads. Bangko Sentral ng Pilipinas (BSP) and industry records indicate Metro Manila condo prices averaged **₱217,000/sqm**, compared to **₱150,400/sqm** in Cavite/Cebu and **₱129,700/sqm** in Laguna—confirming urban premiums were **44% to 67% higher** during the WFH expansion (Mioten, 2024).

```
In [ ]:
```